EZ RACK
SOP - Lath Shelf Repair

I: Objective:

1. To restore all damaged but repairable rack shelves to usable condition by September 30th of each year by:
   a) Reinforcing loose pieces with additional staples.
   b) Using pieces from some shelving to replace missing and damaged pieces on others.
2. To accurately account for the number of repaired shelves moved to and from "repair" status in RTS.

II: Repair Authorization and Invoicing:

- Email Bob Dannenbaum bdannenbaum@ezrack.com to request authorization to repair shelves.
  a) The quantity of repair shelves requested must be equal to or greater than quantity of shelves showing in the repair field in RTS.
  b) An authorization number valid for 60 days.
- Begin repairing only when authorization number is issued.
- When completed, advise Bob Dannenbaum of the completion date and the quantity of repaired shelves moved into WIP and the quantity of shelves moved into scrap.
  a) Invoice EZR for the amount of shelves actually repaired when authorized quantity is completed. The invoice must include the repair authorization number along with the quantity of shelves moved from repair into WIP.
  b) RTS inventory movements in the system must match the total quantity invoiced. EZ needs to see system entries of shelves moved from Repair to WIP, FG, and Scrap.

Example of Internal EZR Tracking Log:

Date Issued	Expiration Date	RC Location	Repair Authorization#	RTS Qty Requested to Start	Paint Color	Invoice Date	Repaired Quantity Invoiced	RTS Shelves Moved from Repair to WIP or Finished Goods	RTS Shelves Moved From Repair to Scrap	Total RTS Qty Transaction	Payment Authorization	Payment Authorization Date	Comments
7/28/2016	9/28/2016	RE Atlanta	REATGA-072816-0001	1200	Red	10/1/2016	1075	1075	125	1200	Y	10/2/2016	EXAMPLE

III: Selecting Shelves to be Repaired or Used for Parts: (Most Important Step)

- During routine repackaging process, all shelving that can be used for parts or be repaired should be moved into the "Repair" field in RTS. The "repair" field should show potential repairable shelves and only include those with loose, missing or damaged pieces.
- Dos and Don'ts of shelf repairing or using as parts:
  - ✓ OSB and plywood shelves: Move directly to scrap (Drivers shouldn't pick these up).
  - ✓ Thin shelves with top or bottom boards less than 3/8" thickness. Move directly to scrap.
  - ✓ Severely weathered shelves that are splitting or have some rot. Move directly to scrap.
  - ✓ Non EZ Rack Shelves that don't have 4 bottom stringers and 10 top boards use as parts or move directly to scrap.
  - ✓ Do not attempt to repair shelves that required replacing three or more top boards. Too difficult. Use remaining boards as parts if possible.
  - ✓ Do not attempt to repair shelves that need bottom stringers replaced. Too difficult. Use shelf for parts if possible.

Examples of repairable versus scrap shelves.

Scrap Shelf: Decent board thickness but too weathered and cracked to repair or use as parts. Move directly to scrap.

Scrap Shelf: Thin wood with top boards or bottom stringers less than 3/8”. Move directly to scrap.

Possible Scrap Shelf: Decent board thickness but too many top boards to replace. Inspect remaining boards for good parts.

Repair: Good board thickness. Loose boards need staples. Inspect all joints. Shoot three staples in any loose joint.

IV: RTS Update Process for Repaired and Scrap Shelves:

- During routine repackaging process, all shelves must be inspected so that shelves that qualify as repairable or can be used for parts are moved into the "Repair" field in RTS. The "repair" field should only show repairable shelves and only include those with loose, missing or damaged pieces.
    - ✓ Refer to section III for QC criteria.
- During the repair process, shelves that are used for parts must be tallied and recorded daily. These counts must then be reported as scrap by making a negative adjustment to "repair" and a positive adjustment to "scrap" on a daily basis. NOTE: shelves used for parts "disappear" and not able to be counted later so an accurate accounting of shelves used for parts must be done when using repair shelves to create a stock of good parts.
- As repaired shelves are completed, the repaired quantities must be moved into WIP daily by making a negative adjustment to "repair" which automatically adds the quantity to WIP. Then later on as good shelves are restacked into finished goods, adjustments to finished goods occurs as normal.
- At the end of the repair project all shelves initially reported as "repair" in RTS and authorized for repair by EZR should be zeroed out. EZR will verify in RTS that the initial quantity has either been moved to repair or scrap. See section II.
  IMPORTANT NOTE: Failure to accurately report repair shelves used as parts as scrap will result in a negative inventory variance during reconciliation and prevent invoice payment authorization.

V: Repair Process:

Tools Required:

- ✓ Air compressor with adjustable pressure gage capable of at least 80 psi.
- ✓ Pneumatic Stapler designed for up to 1.5", 18 gauge staples with ¼" crown. Stapler must have a depth-of-drive adjustment for optimal staple penetration. Brands at Home Depot or Lowes: Hitachi, Bostitch, Porter Cable, Rigid.
- ✓ 1.25" staples, 18 gauge narrow crown, with ¼" crown.

- ✓ Circular saw to cut 48” stringers down to 39.75” top boards.
- ✓ Claw hammer and pliers for removing old staples & nails.
- ✓ Flat head screwdriver for removing old staples.
- ✓ Can of spray paint for color coding. Color to be specified w/ authorization.
- ✓ Work table made by stacking 6-8 pallets.
- ✓ Sheet metal or five+ scrap DRB48s to use under shelves to deflect staples.

If you don’t have a piece of sheet metal to use as a staple deflection base you can use scrap DRB48s instead. Note placement of DRB48s. Depending on pallet top board width some boards may have to be cut with circular saw to create best fit for DRB48 base.

1. Create a work table using a stack of pallets. (see picture above and below)
    a. Make sure the top pallet and stack is very sturdy. Use good pallets.
    b. For a staple deflector base use a piece of sheet metal or lay 5 scrap DRB48 across a top stringer as shown in the picture.

DRB48 hang off pallet sides to allow for space to lay shelf flat on 40” side.

2. Begin by selecting shelves from the repair stacks to use as parts.
    a. Deconstruct enough shelves to create a working stock of shelf parts OR consider purchasing parts if a pallet manufacturer is nearby. Cost of parts could be less than the labor to deconstruct pallets.
    b. Remove or hammer down all old staples and nails.
    c. Cut some 48” stringers down to 39.75” to create more top boards as necessary.

Cut bottom stringers down to 39.75” using circular saw.

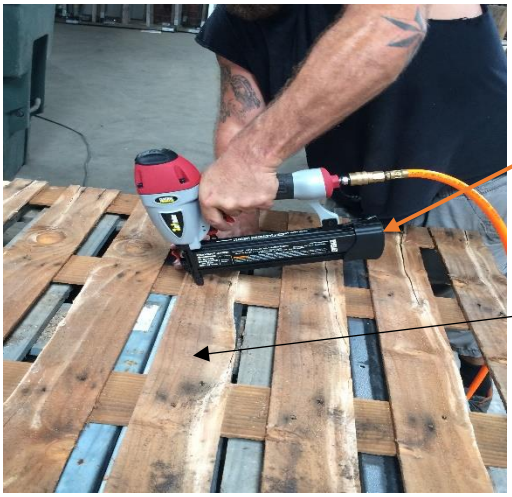
Bottom stringers cut to top board length. You’ll end up with extra bottom stringers as shelves are deconstructed.

3. Set first shelf to be repaired onto table and begin replacing damaged or missing parts.

Place portion of shelf to be repaired on top of DRB48s so that staples will deflect and clinch on bottom. Rotate shelf over DRBs as necessary to repair all joints.

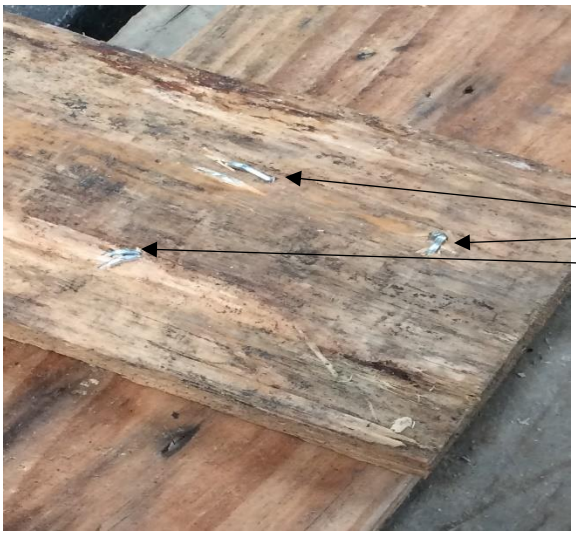
Staple Gun: Adjust depth-of-drive setting on the staple gun to ensure that staple is shot thru top board and stringer, that staple clinches under bottom stringer, and that staple doesn't embed too deeply or not deep enough into top board. The top of the staple should be visible and level with the top board.

4. Shoot three staples at opposing 45 degree angles at each joint to ensure deflection and clinch under the stringer. Always shoot staples at angles.
    - ✓ Always shoot 3 staples per joint.



Work from side of work table so that gun can be maneuvered to shoot staples at opposing angles.

Note shelf placement over DRB48s when repairing middle part of shelf.



Clinched 1.25” staples on bottom. On some thicker shelves, staples may not always clinch but when shot at opposing angles should still be secure enough.

5. Inspect all other joints and shoot additional staples as necessary.
6. Paint a stripe from top to bottom on all 4 sides of every stack of repaired shelves with spray paint before mixing them with WIP and Finished Goods. The Paint color will be specified on the repair authorization.